Mihir Kulkarni

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Education

Columbia University New York, U.S.A.
Ph.D. in Astronomy, Advisor: Prof. Greg L. Bryan 2015–2021

- Thesis: *Modeling the formation, evolution, and observation of first stars*

M.Phil. in Astronomy 2018
M.A. in Astronomy 2017

Indian Institute of Science Education and Research, Pune Pune, India
B.S.-M.S. with distinction in Physics, GPA: 9.1/10 2010–2015

- Thesis: *Simulating the distribution of cosmological neutral hydrogen over cosmic times*

Research interests

- First stars, first galaxies, cosmic dawn, reionization, 21-cm signal, alternative dark matter models, computational astrophysics, semi-analytic modeling.

Experience

Columbia University New York, New York
Deans Fellow Fall 2015 - Summer 2021

The University of Toledo Toledo, Ohio
Postdoctoral research associate Fall 2021 - Summer 2024

Georg-August-Universität Göttingen Göttingen, Germany
Postdoctoral researcher Winter 2024 - Present

Publications

1. **Kulkarni, M.**; Visbal, E.; Bryan, G.L., *Fragmentation in Population III Galaxies Formed through Ionizing Radiation*, 2019, ApJ, 882, 178. (arXiv:1907.11724).

2. **Kulkarni, M.**; Visbal, E.; Bryan, G.L., *The critical dark matter halo mass for Population III star formation: dependence on Lyman-Werner radiation, baryon-dark matter streaming velocity, and redshift*, 2021, ApJ, 917, 40. (arXiv:2010.04169).

3. **Kulkarni, M.**; Ostriker, J.P., *What is the Halo Mass Function in a Fuzzy Dark Matter Cosmology?*, 2022, MNRAS 510, 1425. (arXiv:2011.02116.)

4. Shao, H.; Villaescusa-Navarro, F.; Villanueva-Domingo, P.; Teyssier, R.; Garrison, L. H.; Gatti, M.; Inman, D.; Ni, Y.; Steinwandel, U. P.; **Kulkarni, M**.; Visbal, E.; Bryan, G. L.; Anglés-Alcázar, D.; Castro, T.; Hernández-Martínez, E.; Dolag, K., *Robust Field-level Inference of Cosmological Parameters with Dark Matter Halos*, 2023, ApJ 944, 27. (arXiv:2209.06843.)

5. **Kulkarni, M.**; Visbal, E.; Bryan, G.L.; Li, X., *If Dark Matter is Fuzzy, the First Stars Form in Massive Pancakes*, 2022, ApJL, 941, 18 (arXiv:2210.11515).

6. Shao, H.; de Santi, N. S. M.; Villaescusa-Navarro, F.; Teyssier, R.; Ni, Y.; Anglés-Alcázar, D.; Genel, S.; Hernquist, L.; Steinwandel, U. P.; Castro, T.; Hernández-Martínez, E.; Dolag, K.; Lovell C. C.; Visbal, E.; Garrison, L. H.; **Kulkarni, M.**, *A universal equation to predict Ω_m from halo and galaxy catalogues*, ApJ 956, 149. (arXiv:2302.14591.)

7. Feathers C. R.; **Kulkarni, M.**; Visbal E.; Hazlett R., *A Global Semi-Analytic Model of the First Stars and Galaxies Including Dark Matter Halo Merger Histories*, ApJ, 962, 62. (arXiv:2306.07371).

8. Hazlett R.; **Kulkarni, M.** Visbal E.; Wise, J. H., *A Framework to Calibrate a Semianalytic Model of the First Stars and Galaxies to the Renaissance Simulations*, 2025. ApJ, 978, 13. (arXiv:2403.05624).

9. Feathers C. R.; **Kulkarni, M.**; Visbal E., *From dark matter minihalos to large-scale radiative feedback: a self-consistent 3D simulation of the first stars and galaxies using neural networks*, 2025. JACP, 02, 43. (arXiv:2411.07875).

10. Sullivan, J.; Haiman, Z.; **Kulkarni, M.**; Visbal E., *Can supermassive stars form in protogalaxies due to internal Lyman-Werner feedback?*, 2025, MNRAS, 542, 822. (arXiv:2501.12986).

11. Ambrose, A. E.; Visbal, E.; **Kulkarni, M.**; McQuinn, M., *Radiative Transfer Simulations of Ly Intensity Mapping During Cosmic Reionization Including Sources from Galaxies and the Intergalactic Medium*, 2025. JACP, 08, 080. (arXiv:2502.18654).

12. Behling, T.; Hazlett, R.; **Kulkarni, M.**; Visbal, E., *Evaluating the Accuracy of Reionization Prescriptions in Semi-analytic Models of the First Stars and Galaxies* , 2025 (arXiv:2508.04808).

13. Hazlett, R.; Mead, J.; Visbal, E.; Bryan, G. L.; Mac Low, M.; **Kulkarni, M.**; Andersson, E. P.; Brauer, K.; Wise, J., *From Primordial Stars to Early Galaxies: A Semi-Analytic Model Calibrated with Aeos and Renaissance* , 2025, submitted to ApJ (arXiv:2510.11629).

Fellowships and Awards

- **Dean's Fellowship** at Columbia University. 2015–2021
- **Junior Research Fellowship** (JRF - NET) of Council of Scientific and Industrial Research (CSIR), Govt. of India with an All India Rank of 25. 2013
- **Innovation in Science Pursuit for Inspired Research** (INSPIRE), Department of Science and Technology, Govt. of India. 2010–2015
- **National Talent Search Examination** (NTSE) Scholarship, National Council of Education Research and Training (NCERT), India. 2008

Presentations

- *Simulations and semi-analytic models of the first stars and galaxies*, **HERA2025: The Physics of Galaxies at the Epoch of Reionization**, Pisa, Italy, September 2025 (talk).
- *Simulations and semi-analytic models of the first stars and galaxies*, **Sapienza University of Rome**, Rome, Italy, September 2025 (seminar).
- *Role of the halo accretion history in the formation of Population III stars*, **First Stars VII**, New York, U.S.A., May 2024 (lightning talk).
- *Hydrodynamical simulations of the first stars in the cold and fuzzy dark matter cosmologies*, **HI as a cosmological probe**, Nazareth, Israel, May 2023 (talk).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **MIT Computational Structure and Galaxy Formation group meeting**, May 2023 (seminar).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **Columbia University**, New York, May 2023 (seminar).

- *Hydrodynamical simulations of the first stars in the cold and fuzzy dark matter cosmologies*, **Fermilab**, Illinois, May 2023 (seminar).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **University of Toledo**, Ohio., February 2023 (**Colloquium**).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **Tata Institute of Fundamental Research (TIFR)**, Mumbai, India, December 2022 (seminar).
- *If dark matter is fuzzy, the first stars form in massive pancakes*, **Inter-University Center for Astronomy and Astrophysics (IUCAA)**, Pune, India, December 2022 (seminar).
- *Formation of the first stars and galaxies in a fuzzy dark matter cosmology*, **AAS 240th meeting**, July 2022 (talk).
- *Population III stars and processes that delay their formation*, **AAS 237th meeting**, January 2021 (dissertation talk).
- *Population III star formation: effects of UV radiation, baryon-dark matter streaming velocity, and redshift*, **Galaxies discussion group, University of Cambridge**, October 2020 (talk).
- *The critical dark matter halo mass for Population III star formation: dependence on Lyman-Werner radiation, baryon-dark matter streaming velocity, and redshift*, **The First Stars, SAZERAC**, October 2020 (talk).
- *A critical mass for Pop III stars: dependence on LW radiation, dark matter-baryon streaming and redshift*, **SAZERAC**, July 2020 (poster).
- *A critical mass for Pop III stars: dependence on LW radiation, dark matter-baryon streaming and redshift*, **First Stars VI**, Concepción, Chile, March 2020 (poster).
- *A critical mass for Pop III stars: dependence on Lyman-Werner radiation, baryon/dark-matter streaming, and redshift*, **AAS 235th meeting**, Honolulu, Hawaii, January 2020 (talk).
- *Fragmentation in Ionized Pop III Galaxies*, **Into the Starlight: The End of Cosmic Dark Ages**, Aspen, Colorado, March 2019 (talk).
- *Fragmentation in Ionized Pop III Galaxies*, **Cosmology: the Next Decade**, International Centre for Theoretical Sciences, Bengaluru, India, January 2019 (talk).
- *Fragmentation in Ionized Pop III Galaxies*, **Astrophysical Frontiers in the Next Decade and Beyond**, Portland, Oregon, June 2018 (poster).
- *Fragmentation in Ionized Pop III Galaxies*, **Enzo workshop**, Atlanta, Georgia, May 2018 (talk).
- *Commissioning of a new 15-m radio telescope at NCRA*, **Astronomical Society of India**'s annual meeting, Pune, India, February 2015 (poster).

Teaching

- **Co-Instructor** at Georg-August-Universität Göttingen — Summer 2026
  The First Billion Years
- **Teaching Assistant** at Georg-August-Universität Göttingen — Summer 2025
  Introduction to Cosmology
- **Teaching Assistant** at Georg-August-Universität Göttingen — Winter 2024 and 2025
  Introduction to General Relativity
- **Guest lectures** at the University of Toledo — Spring 2021-Fall 2021
  on numerical methods in the class Computational Physics
- **Guest lectures** at the University of Toledo — Fall 2021
  on Planetary Geology in the class Solar System Astronomy
- **Co-Instructor** at Columbia University — Fall 2017 - Spring 2020
  Modern Cosmology course for the Science Honors Program for high school students
- **Teaching Assistant** at Columbia University — Spring 2018
  Modeling the Universe.
- **Instructor** at Columbia University — Fall 2017

Astronomy Lab 1: Earth, Moon and Planets.

- **Observational Teaching Assistant** at Columbia University — Spring 2017
  Setting up and helping with the observing sections of Astronomy labs.
- **Instructor** at Columbia University — Fall 2016
  Astronomy Lab 2: Stars, Galaxies and Cosmology.

Mentoring

- **Li Ma** Ph.D. student at the University of Göttingen. — Fall 2024 –present
- **Colton Feathers** Ph.D. student at the University of Toledo. — Fall 2021–Summer 2024
- **Ryan Hazlett** Ph.D. student at the University of Toledo. — Fall 2021–Summer 2024
- **Abigail Ambrose** Ph.D. student at the University of Toledo. — Fall 2021–Summer 2024
- **James Sullivan** Ph.D. student at Columbia University. — Fall 2022–Summer 2024
- **Julian Kleff** Masters thesis student at the University of Göttingen. — Summer 2025 –present
- **Monja Begau** Masters thesis student at the University of Göttingen. — Summer 2025 –present
- **Jack King** Masters thesis student at the University of Göttingen. — Summer 2026 –present
- **Jinhwan Jeon** Bachelor thesis student at the University of Göttingen. — Summer 2026 –present
- **Ivana Hildebrandt** Bachelor thesis student at the University of Göttingen. — Summer 2025 –Winter 2025
- **Thomas Behling** Undergraduate student at the University of Toledo. — Summer 2023–Summer 2024

Public Outreach

- Public Outreach talk on 'Cosmic Dawn: First Stars and Galaxies' — November 2025
  As part of the outreach lecture series at the Institute for Astrophysics and Geophysics in Göttingen.
- Public Outreach talk on 'Clocks of the Universe' — July 2025
  As part of Nacht der Wissen (Night of Knowledge) in Göttingen.
- Public Outreach talk on 'Clocks of the Universe' — September 2018
  A part of the Columbia astronomy outreach lectures series. Covered in press 1, 2.
- Regular volunteer for Columbia astronomy outreach events — 2015–2018
  Night sky observations using telescopes.
- Regular volunteer for Rooftop Variables at Columbia University — 2015–2021
  A group for interacting with high school astronomy clubs in the New York City area.
- Volunteer at Reading Team Math — 2017–2018
  After-school program for math education for young children.

Skills

- **Languages:** Python (13 years), Cython (7 years), C (7 years).
- **Tools:** ENZO, GADGET-2, MUSIC, YT, YTREE.

Github profile

References

1. Prof. Greg L. Bryan
   Department of Astronomy,
   Columbia University,
   New York, NY, U.S.A.
   email: gbryan@astro.columbia.edu

2. Prof. Eli Visbal
   Department of Physics and Astronomy,
   University of Toledo,
   Toledo, OH, U.S.A.
   email: elijah.visbal@utoledo.edu

3. Prof. Jens Niemeyer
   Institut für Astrophysik und Geophysik,
   Georg-August-Universität Göttingen,
   Göttingen, Germany.
   email: jens.niemeyer@phys.uni-goettingen.de

4. Prof. Zoltan Haiman
   Department of Astronomy,
   Columbia University,
   New York, NY, U.S.A.
   email: zoltan@astro.columbia.edu